

Tian Yi Lim

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I am a driven and resourceful Robotics Autonomy Engineer with a focus on finding localisation and perception solutions.
I take pride in using the appropriate mix of cutting-edge and proven algorithms to build robust, useful robots.

PROFESSIONAL EXPERIENCE

DSO National Laboratories

Jan. 2025 – Present

Member of Technical Staff – Robotics Division

Singapore

I work on localization and perception tasks for various robotic platforms on the ground and air, using a broad variety of sensing modalities and algorithms. Some of these projects include:

- Developed a stereo depth perception algorithm for Event Cameras for drones, enabling fast autonomous flight in unmapped environments, even in low-light conditions.
- Implemented LiDAR-Inertial Odometry algorithms for robust localisation, cm-level drift in 100-m scale environments.
- Visual Positioning System for GNSS-independent global localisation.
- Online 3D Scene Graph construction for metric-semantic mapping of indoor environments.
- **Tools used:** C++, DDS, Python, PyTorch, TensorRT | Jetson Orin, Ouster LiDAR, VectorNav IMUs, ...

Embotech AG

Aug. 2023 – Feb. 2024

Perception Intern – Safety and Systems Engineering Team

Zurich, Switzerland

I helped implement a LiDAR-based perception stack for L4 autonomous trucks. My work was a key part of a proof of concept autonomy solution, which has since been successfully trialled in multiple ports in Europe.

- Implemented a geometric clustering and tracking algorithm for obstacle detection.
- Architected sensor fusion rules for multimodal perception inputs.

EDUCATION

ETH Zurich

Sep. 2022 – Nov. 2024

MSc in Robotics, Systems and Control

Zurich, Switzerland

Spent two years learning robotics fundamentals from some of the best professors in the field.

Put theory into practice with various practical courses e.g. with RL-based drones and autonomous racecars.

- **Semester Thesis:** System Lead for F1TENTH autonomous racecar from 2023-2024. Over three global-level competitions, the team won 1st once and 3rd twice.
- **Masters Thesis:** Two-view loop closure constraints for SLAM, supervised by Hermann Blum and Marc Pollefeys. Work published in RA-L.
- **Relevant Modules:** Planning and Decision-Making for Autonomous Robots, Vision Algorithms for Mobile Robots, Vision-based Drone Flight, Computational Control

Imperial College London

Sep. 2019 – Jun. 2022

BEng in Electronic and Information Engineering

London, UK

- Graduated in the **top 10% of cohort**
- **Final Year Project:** Multi-Robot Path Planning, supervised by Andrew Davison
- **Relevant Modules:** Robotics, Robotic Manipulation, Computer Vision, Intro to Machine Learning

PUBLICATIONS

Robotics and Automation Letters

2026 | First Author

Loop Closure From Two Views: Revisiting PGO for Scalable Trajectory Estimation Through Monocular Priors

Journal of Field Robotics

2024 | Contributor

ForzaETH Race Stack – Scaled Autonomous Head-to-Head Racing on Fully Commercial Off-the-Shelf Hardware